DARK ENERGY IS PURE QUANTIZED EXISTENTIAL ENERGY

A Foundational Declaration Distinguishing Dark Energy from Vacuum Energy in Physical Fields

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Introduction

This paper presents a foundational declaration within the framework of the Dahli Model, proposing a new definition of dark energy in terms of its existential origin, quantum nature, and cosmic structure.

The declaration is based on a fundamental principle: existence precedes the universes, and pure existential energy is the sum of the pure cosmic energies of all universes, without any activation. This energy takes a quantized structure, such that each existential quantum represents one pure cosmic energy.

Based on this conception, the Dahli Model establishes that dark energy is pure quantized existential energy, and is neither the vacuum energy contained in physical fields nor the minimum energy of those fields.

The model establishes a distinction between two types of cosmic vacuum: the relative vacuum associated with physical fields, and the pure cosmic vacuum resulting from the superposition of fields, which is most likely to take the form of a brane in an uncontracted pure cosmic spatial dimension, to which the existential quanta are connected through contracted pure cosmic temporal dimensions.

Through this distinction, the Dahli Model separates vacuum energy, which is associated with the pure counterpart of our universe, from dark energy, which represents the remaining quantized pure cosmic energies. It also establishes the relationship between the depletion of existential quanta and the evolution of cosmic expansion within its theoretical framework.

This paper aims to announce this definition, identify its existential, energetic, and geometric principles, and present a clear conceptual structure that distinguishes dark energy from vacuum energy, while establishing a pathway for its mathematical and physical formulation and the testing of its predictions.

Keywords: dark energy; pure quantized existential energy; Dahli Model; existential quanta; pure cosmic vacuum; brane; contracted temporal dimensions; vacuum energy; cosmological constant problem; cosmic expansion; multiverse

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The Declaration

Dark energy is not the vacuum energy in physical fields.

On this date, November 4, 2026, we declare that we are the first to put forward this foundational declaration within the framework of the Dahli Model, based on a new definition of dark energy, its

existential origin, and its cosmic structure:

Dark energy is pure quantized existential energy.

Dark energy is not field energy, it is not the vacuum energy contained in physical fields, and it is not merely a mathematical constant in quantum field theory.

1. What Is Pure Existential Energy?

Pure existential energy is the sum of the pure cosmic energies of all universes, without any activation.

Existence precedes the universes, just as the universe precedes galaxies.

Therefore, pure existential energy does not belong to a single universe; rather, it encompasses the sum of the pure energies of all universes.

- **Existential:** Because it belongs to the level of total existence that precedes the universes.
- **Pure:** Because it is the sum of the pure energies of all universes, without any activation.
- **Quantized:** Because it consists of multiple existential quanta, each of which represents one pure, non-activated cosmic energy.

Dark energy is this same pure quantized existential energy; that is, it is the sum of the quanta that represent the pure cosmic energies of all universes. These quanta are the dark energy existing in our pure cosmic vacuum, whose nature is most likely a brane in the single pure spatial dimension of our universe.

2. Where Does Pure Existential Energy Exist?

All existential quanta are connected to the pure cosmic vacuum of our universe, with the exception of one cosmic energy that is exactly the counterpart of our universe's pure vacuum. It is quantized into what is called vacuum energy, because the two counterparts—our pure cosmic vacuum and our pure cosmic energy—do not meet, thereby quantizing it. Our quantized pure cosmic energy spreads through the fields in what is called field energy, i.e. the known vacuum energy, namely the energy that has been measured theoretically.

The pure cosmic vacuum is not an independent physical field. Rather, it results from the superposition of relative fields and the superposition of fields in a single, uncontracted pure cosmic spatial dimension, which is likely to give rise to a brane. This is the same pure cosmic vacuum that physicists have overlooked, and whose neglect was the cause of the cosmological constant problem.

This brane requires contracted pure cosmic temporal dimensions, with one existential quantum existing in each of them, i.e. one pure, non-activated cosmic energy. This means that every existential quantum, i.e. every pure cosmic energy, exists in a contracted cosmic temporal dimension belonging to that cosmic energy. These contracted temporal dimensions do not belong to our universe; rather, they are connected to our pure cosmic vacuum spatially only, not spatially and temporally.

Accordingly:

- The pure cosmic vacuum is the brane itself.
- The brane exists in one uncontracted pure cosmic spatial dimension.
- Its temporal dimensions are contracted pure cosmic temporal dimensions.

- Each contracted pure cosmic temporal dimension contains one existential quantum representing one pure cosmic energy.

This brane is not a string; rather, it is more likely to be a brane.

Therefore, dark energy is not something separate from this pure vacuum of our universe; it is the existential quanta themselves, i.e. the pure cosmic energies existing in the brane that represents our pure cosmic vacuum. Rather, it is existential energy that is spatially connected and temporally separate from our pure cosmic vacuum, and it has no relation to fields or to the minimum energy of fields, i.e. it has no relation to vacuum energy.

3. There Are Two Types of Cosmic Vacuum

The Dahli Model establishes the existence of two different types of cosmic vacuum:

First: The Relative Vacuum — The Field Vacuum

It is the vacuum associated with relative physical fields, in which the minimum energy of these fields exists, i.e. their own vacuum energy. This is itself the pure cosmic energy that is exactly the counterpart of our pure cosmic vacuum, i.e. the pure energy of our universe. It is quantized and decomposed into what is called vacuum energy.

Second: The Pure Cosmic Vacuum

It results from the superposition of relative fields and is not an independent physical field. It is itself the pure cosmic spatial brane, existing in one uncontracted pure cosmic spatial dimension, with contracted pure cosmic temporal dimensions in which the existential energy quanta exist, i.e. the pure cosmic energies, with the exception of the pure energy of our universe, which is quantized by virtue of its symmetry with the pure vacuum of our universe, i.e. the brane, and is quantized and distributed across the fields under the name of the minimum energy of the fields, i.e. vacuum energy.

4. Vacuum Energy and Dark Energy: Two Different Energies

The Dahli Model distinguishes between two energies that differ in origin, nature, and affiliation:

Vacuum Energy in Fields	Dark Energy
The minimum energy of physical fields	Pure existential energy
Represents the counterpart of the pure vacuum, i.e. our pure cosmic energy	Represents the sum of the pure energies of all universes, except our pure cosmic energy
Quantized and decomposed into vacuum energy, i.e. field energy	Quantized into existential quanta
Belongs to the pure counterpart of our cosmic vacuum	Includes the pure energies of all universes
Associated with the field vacuum	Exists in the pure cosmic vacuum, i.e. in the brane
Is not dark energy	Is itself pure quantized existential energy

Accordingly, vacuum energy in the fields is not dark energy, and the two must not be confused.

5. The Depletion of Existential Energy and Its Relationship to Cosmic Expansion

The Dahli Model states that the pure cosmic vacuum depletes the pure quantized existential energy contained within it; that is, it depletes the quanta of pure, non-activated cosmic energies that represent dark energy.

This depletion results in a gradual decrease in the number of existential quanta present in the brane.

The more the number of existential quanta decreases, i.e. the more dark energy is depleted, the greater the expansion of the universe becomes.

Thus, cosmic expansion in the Dahli Model is linked to the depletion of the reservoir of pure quantized existential energy, and the decrease in its quanta is associated with an increase in cosmic expansion.

6. Foundational Conclusions

This declaration is based on the following principles:

1. Existence precedes the universes, just as the universes precede galaxies.
2. Pure existential energy is the sum of the pure energies of all universes, without any activation.
3. Cosmic vacuum is of two types: a relative vacuum of fields, and a pure cosmic vacuum resulting from the superposition of fields.
4. The pure cosmic vacuum itself is a brane in one uncontracted pure cosmic spatial dimension.
5. The temporal dimensions of this brane are contracted pure cosmic temporal dimensions, with one existential quantum existing in each of them.
6. Dark energy is itself the pure quantized existential energy existing in the brane.
7. The pure cosmic vacuum, i.e. the brane, depletes this quantized existential energy, and the more its quanta decrease, the greater the expansion of the universe becomes.

Final Declaration

Dark energy is pure quantized existential energy: the sum of the pure cosmic energies of all universes, without any activation. It consists of existential quanta, each of which represents one pure cosmic energy and exists in a contracted pure cosmic temporal dimension. All these quanta are connected to the pure cosmic vacuum of our universe, with the exception of the pure energy of our universe, which is quantized and distributed across the fields, i.e. it is vacuum energy.

The pure cosmic vacuum that contains dark energy, i.e. quantized existential energy, is the superposition of all fields in a pure cosmic spatial dimension. The result will not be a superposed field; rather, the result is more likely to be a brane. This brane exists in the single pure spatial dimension of our universe. This brane depletes the quanta of existential energy contained within it, and the more these quanta decrease, the greater the expansion of the universe becomes.

This is the foundational definition of dark energy in the Dahli Model.

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